#7 · 7.1% of marks

Ch 11 — International Financial Management

SM TYK theory Qs: **instruments of international finance** · **impact of GDRs on Indian capital market**. Numericals: foreign project NPV (PPP / IRP rates, remittance, double tax), APV, GDR number & cost, ungear / regear beta, netting.

① International capital budgeting — complexities (SM 1.1)

1. Foreign cash flows must be **converted** into parent's currency
2. **Parent** cash flows differ from **project** cash flows
3. Remitted profits taxed in **home and host** country
4. Effect of **forex risk** on parent's cash flow
5. Changing **inflation** shifts the competitive environment
6. Host-country **restrictions on distribution** of cash
7. Initial investment in host country to benefit from release of **blocked funds**
8. **Political risk** reduces chance of expected cash flows
9. Host-country **concessions / benefits** raise profitability
10. **Terminal value** hard to estimate (buyers have divergent views)

② Problems in foreign investment analysis (SM 1.2)

→ **Forex risk**: forecast host inflation; inflate local-currency cash flows; convert at spot × expected depreciation from **PPP**

→ **Restrictions** on profit transfer, depreciation charges, technical differences → project ≠ parent flows; diluted via **internal transfer prices, overhead payments**; adjust blocked funds at their **opportunity cost**

→ **Taxes**: two regimes, royalties / dividends / management fees, withholding tax, treaties, discrimination between remitted vs re-invested profit. Use **higher of home / host tax rate**; if feasible, acceptable under a more favourable regime; if not, look for other tax savings. MNCs cut tax via **transfer pricing**.

③ Project vs parent cash flows (SM 1.3)

→ **Project's own cash flows**: must beat local competitors (else invest in local equity / bonds) — but foreign projects often **replace imports**, so comparison is hard; local evaluation **avoids currency conversion**

→ **Parent's view**: operating + financial flows **actually remitted** are the yardstick (dividends, debt service). Evaluate on **net after-tax operating cash flows**; don't mix financial flows into operating flows.

④ Discount rate vs adjusting cash flows (SM 1.4)

→ Earlier MNCs **raised the discount rate** for political & FX risk. SM: not proper to combine all risks in one rate — political risk relates to future events that can't be foreseen now, so (SM's words) adjusting the rate for it "**penalises early cash flows more than distant cash flows**"; raising it for FX risk ignores **favourable** currency moves.

→ Better: **adjust cash flows** — convert remitted flows into **certainty equivalents** (probability + decision tree), discount at host or parent rate.

NUMERICAL CORE

5 Adjusted Present Value (SM 1.5)

$$APV = -I_0 + \Sigma X_t \div (1 + k^*)^t + \Sigma T_t \div (1 + i_d)^t + \Sigma S_t \div (1 + i_d)^t$$

k^* = **unlevered** cost of capital (operating CF) · T_t = tax saving from financing mix · S_t = before-tax interest **subsidies** · i_d = **before-tax cost of debt** (for the last two terms)

Value-additive: each component discounted at a rate matching its risk. Test basic viability first; if feasible stop, else add the other complexities.

Blocked funds: balances abroad that can't be remitted — **direct** (exchange control) or **indirect** (local / withholding taxes on remittance). If released for the project, they **reduce the investment outlay**.

May 2026 Q3(a) (ICAI): pick cheapest borrowing (forward by IRP), then APV with tax shield discounted at cost of debt.

6 SM scenarios (1.6) — method & answers

Scenario	How SM solves it
Illus 1 — foreign co. (Perfect Inc., US) invests in India	Exchange rates by **PPP**: $E(S_1) = S_0 \times (1 + I_d) \div (1 + I_f)$; inflate price; 55% revenue share + royalty; Indian tax 30% paid next year; only **50% remittable** in year earned, rest next year; convert remittance to $; US tax 35% with DTAA credit (net = 35% − Indian tax), paid next year; discount at **15%** (not WACC 13%, project riskier) → NPV **−$4.15 mn** → reject
Illus 2 — Indian co. invests in Nepal, funds raised there	Use **competitor's** beta: ungear 1.35 × 1850 ÷ [1850 + 510(1 − 0.20)] = **1.106**; regear at 55 : 45 → equity β **1.83**; K_e = 8 + 1.83 × 3 = **13.49%**; WACC = 13.49 × 0.55 + 9 × 0.8 × 0.45 = **10.66%**. Sunk research cost and apportioned overheads ignored → reject
Illus 3 — Indian co. invests in China via **GDR** (Mauritius)	K_e (GDR) = 2.75 ÷ 198 + 10% = **11.39%**; rates by **IRP**: 9.50 × 1.12/1.10 = **9.67**, × (1.12/1.10)² = **9.85**; NPV in CN¥ **−12,025**; in ₹ at 12%: remitted yearly **+₹8,70,008**, remitted at end **+ ₹2,14,257** → accept (₹ weakens vs CN¥)

TRAP

Illus 1: remittance and tax both lag a year → there are year-6 and year-7 cash flows. Illus 3: SM discounts CN¥ flows at 11.39% but ₹ flows at 12%, which is why the two NPVs differ in sign — reproduce SM's working as given.

TYK 1 — foreign currency approach: find the foreign-currency discount rate via IRP, (1.14 × 1.08 ÷ 1.12) − 1, discount $ flows, convert NPV at spot. **Home currency approach**: convert flows at IRP forward rates, discount at 14%.

⑦ FCCBs (SM 2.1)

*Convertible bond in a currency other than issuer's; regular coupon + principal, with option to convert into stock (via warrants activated at a price). Equity feature → **lower coupon**.*

Advantages

*(i) investor flexibility — convert or redeem (normally **3 years**) · (ii) **delayed dilution** of equity / EPS · (iii) easily marketable — conversion upside + minimum fixed interest*

Disadvantages

*(i) **exchange risk** on interest → only low D/E, large forex earners opt · (ii) more debt + forex outgo · (iii) low rate (3–4%) but FX risk on interest & repayment if not converted; benefit only if growth lets shares be issued above current price*

⑧ ADRs & GDRs (SM 2.2–2.3)

- → ***ADR**: issued in USA under stringent **SEC** rules; non-US company's shares deposited with custodian in home country; US$-denominated, traded like US securities; same rights as underlying holders*
- → ***GDR**: negotiable certificate in a currency not native to issuer, representing local-currency equity; mostly USD (some Euro / £); can trade in **more than one currency**, inside and outside US. Shares delivered to custodian → depository (eg Bank of New York) issues DRs. Indian issues took GDR form — marketed globally; fetched higher prices than domestic issues; no other term forex funding available.*

Impact on Indian market (TYK Q2)

*(i) market shifting from **Bombay to Luxembourg** (ii) **arbitrage** possible (iii) market no longer independent of the world (iv) **retail investors sidelined** — free pricing ends discounted issues. Large foreign inflow.*

Markets

*(i) mainly **institutional** investors (ii) emerging-market funds dominate (iii) FIIs switch from shares to GDRs (iv) UK, USA (QIBs), SE Asia (HK, Singapore), France, Switzerland*

***Characteristics**: (i) economic benefits but **no voting rights** (ii) settled via **CEDEL & Euro-clear** (iii) listed on **Luxembourg SE** (iv) OTC trading by market makers (v) freely traded (vi) marketed globally, multi-currency (vii) dividend in issuer currency converted to $ → **FX risk with investor** (viii) cancel after **45-day cooling period**; cost = BSE / NSE price on redemption-advice date.*

***Illus 4**: need $10 m net, 2% float → gross $10.204 m; GDR = 2 sh × 250 × 90% = ₹450 = $7.50 → **1.3605 m GDRs**; K_e = 40 ÷ 441 + 12% = **21.07%**.*

⑨ Euro-convertible bonds & others (SM 2.4–2.5)

- → ***ECB**: convert into predetermined shares (conversion premium), fixed interest, interest & redemption in US$. **Call** (issuer) — redeem early, forces conversion when share price **~130–150%** of conversion price. **Put** (holder) — sell back at set price & date.*
- → ***Euro-convertible zero coupon** (no interest, converts at maturity, ~5 yrs, deferred equity) · **Euro-bonds with equity warrants** (market coupon, detachable warrants) · **Syndicated bank loans** (5–10 yrs, SOFR + spread, covenants) · **Euro-bonds** (currency issued outside its country, eg Yen bond in France; tax / regulation refuge; bearer — fixed, FRN, convertible) · **Foreign bonds** (in lender country's currency, sold there, restricted) · **Euro CP** (< 1 year, discount) · **Credit instruments** (rates differ by speed / risk)*

10 International Financial Centre (SM 3)

IFC caters to customers **outside its own jurisdiction** — hub for funds, products, services on own land under a **different set of regulations**; flexibility in currency trading, insurance, banking.

Benefits (5)

(i) professionals abroad can come and practise here (ii) global opportunities **without leaving homeland** (iii) stops **brain drain** (iv) brings back transactions now done abroad (v) complex derivatives can be traded from India

Constituents (6)

(i) highly developed **infrastructure** (ii) stable **political** environment (iii) **strategic location** (airport, seaport, weather) (iv) **quality of life** (v) rational, fair, transparent **regulation** (vi) **sustainable economy** absorbing shocks

- **GIFT City**: idea 2007 (rival Dubai, Hong Kong) — retain financial business moving out of India. IFSC operationalised at GIFT Multi Services SEZ in **April 2015**; competitive tax regime in **Budget 2016**
- **India INX** (BSE subsidiary) inaugurated **9 Jan 2017**; **NSE** began at GIFT **5 June 2017** (equity, currency, IRF, commodity derivatives)
- Low cost, competitive tax, single-window clearance, relaxed company law, international arbitration centre; moving to unified regulation. **FEMA not applicable** at GIFT City.

11 Sovereign wealth funds (SM 4)

State-owned fund of government money, usually from surplus reserves; legal basis varies (constitutive law, fiscal law, constitution, company law, others).

Sources (5)	Surplus from state natural-resource revenues & trade surpluses · bank reserves from budget excesses · foreign currency operations · privatisation money · government transfer payments
Objectives (8)	Protect & stabilise budget / economy from revenue volatility · diversify from non-renewable commodity exports · beat returns on forex reserves · help monetary authority absorb unwanted liquidity · savings for future generations · fund social & economic development · sustainable long-term growth for target countries · political strategy
Classifications (5)	Stabilisation · savings / future generation · public benefit pension reserve · reserve investment · strategic development SWF (SDSWF)
Sovereign vehicles (4)	SWFs · public pension funds · state-owned enterprises · sovereign wealth enterprises (SWEs)

Risk management ranges from very conservative to high risk tolerance, depending on assets and objectives.

12 Why international WC is complex (SM 5)

(1) wider financing options — local or global (2) interest & tax rate differentials (3) **forex risk** on flows and trade (4) home / host **restrictions** on cash & inventory movement (5) limited knowledge of host conditions (6) only with **full capital convertibility** can funds be positioned freely (not India).

13 Multinational cash management (SM 5.1)

→ **Traditional objectives**: (1) manage & control cash (2) optimum utilisation & conservation of funds

→ **Effective system objectives (5)**: minimise currency exposure · minimise overall cash need · minimise transaction costs · minimise political risk · economies of scale & superior knowledge

→ **Centralised system helps**: (a) minimum cash balance (b) manage centre's liquidity (c) optimal hedging (d) max returns on cash (e) **multinational netting** (f) **transfer pricing** (g) exploit currency correlations — receivables & payables with positive correlation; payables with negative correlation; pooling → **group variance < sum of individual variances**

→ **Two basic objectives**: optimise cash flow movements + invest excess cash. **Six ways**: accelerate inflows (lockboxes, pre-authorised payment) · manage blocked funds (local bank finance, transfer pricing, negotiation, leads / lags) · **leading & lagging** · transfer pricing · **netting** · invest excess cash (Euro-currency market, pooled deposits; weigh conversion costs)

Leading / lagging: adjust timing of intra-group payments on currency expectations — **pay hard-currency payables early (lead)**, **delay soft-currency payables (lag)**. Indian subsidiary paying Japanese one: expects ₹ to fall vs ¥ → **lead**; expects ₹ to strengthen → **lag**. Check host-country restrictions.

14 Netting (SM 5.1.5)

Advantages: fewer cross-border transfers (admin cost) · less FX conversion (transaction cost) · better cash forecasting (net transfers at period end) · accurate reporting and coordinated settlement.

→ **Bilateral**: X buys $20 m from Y, Y buys $30 m from X → Y pays X **$10 m** only (not $50 m)

→ **Multilateral**: each affiliate nets all receipts against all payments; needs centralised communication + disciplined subsidiaries. P → Q → R → P, $50 m each → transfers **fully eliminated**

SM matrix ($000)	Receipts	Payments	Net
France	200	240	pays **40**
Germany	180	130	receives **50**
UK	210	160	receives **50**
Italy	190	250	pays **60**

Gross $780 k → net **$100 k = 87%** reduction.

15 International inventory (SM 5.2)

MNCs hold more than EOQ — **stock piling** — due to import bottlenecks (political) and expected **depreciation** of the importing country's currency. Decide by weighing **cumulative carrying cost vs expected input price rise**; if supply interruption is very likely, stockpile even if costly. Longer lead times + customs → **re-order point much earlier**; reliance on imports largest in a **vertical** set-up.

16 International receivables (SM 5.3)

- → **Inter-firm sales**: exporter wants a **strong** currency, importer a **weak** one; exporter may invoice in a weak currency if it has **debt in that currency** (natural offset). Weak-currency invoice → short credit; liberal credit lets exporter borrow against bills, or is forced by competition (financing importer's inventory) → more sales.
- → **Intra-firm sales**: global allocation of resources; timing doesn't matter to the group — cash-rich unit pays early, cash-short pays late → **leads & lags**.

17 ICAI exam patterns (from papers; app-written summary)

- → **Currency swap inside a project** (May 2026 Q1(a)): principal exchanged at the inception spot rate both ways — no FX risk on principal; other FC flows at forward rates
- → **Uncertain FC flows** (RTP May 2025): CFAT for each exchange-rate scenario → probability-weighted CFAT → perpetuity CFAT(1 + g) ÷ (k − g) − outlay
- → **GDR number & cost** (PYQ May 2024): gross issue = net ÷ (1 − flotation) — same steps as SM Illus 4

18 Correlation — how Ch 11 actually connects

Ch 11 idea	The actual link (app-written cross-reference; the Ch 11 side is SM)
PPP / IRP forecast rates	**Ch 10**: same formulas — PPP $E(S_1) = S_0(1 + I_d)/(1 + I_f)$ (Illus 1); IRP $F = S(1 + r_h)/(1 + r_f)$ (Illus 3). Direct quote: home inflation / rate on top.
Ungear / regear beta	**Ch 6 / Ch 3**: $\beta_{asset} = \beta_{equity} \times E \div [E + D(1 - t)]$ (debt $\beta = 0$); use a **proxy** firm in the same business; regear at the project's own D/E; then CAPM → WACC.
APV	**Ch 3**: base-case NPV at unlevered k + PV of financing side-effects; tax shields at cost of debt — identical method, just with FX flows.
GDR cost $K_e = D_1 \div$ net proceeds + g	**Ch 5**: Gordon growth model; flotation cost lowers net proceeds and raises K_e.
FCCB / ECB call & put	**Ch 9 / Ch 5**: convertible = straight bond + call on shares; issuer's call forces conversion once share price is well above conversion price.
Leading & lagging, netting	**Ch 10**: internal hedging techniques — same definitions; netting cuts transaction exposure before any external hedge.
Pooling cash	**Ch 6**: less-than-perfectly correlated flows → group variance < sum of individual variances (diversification).